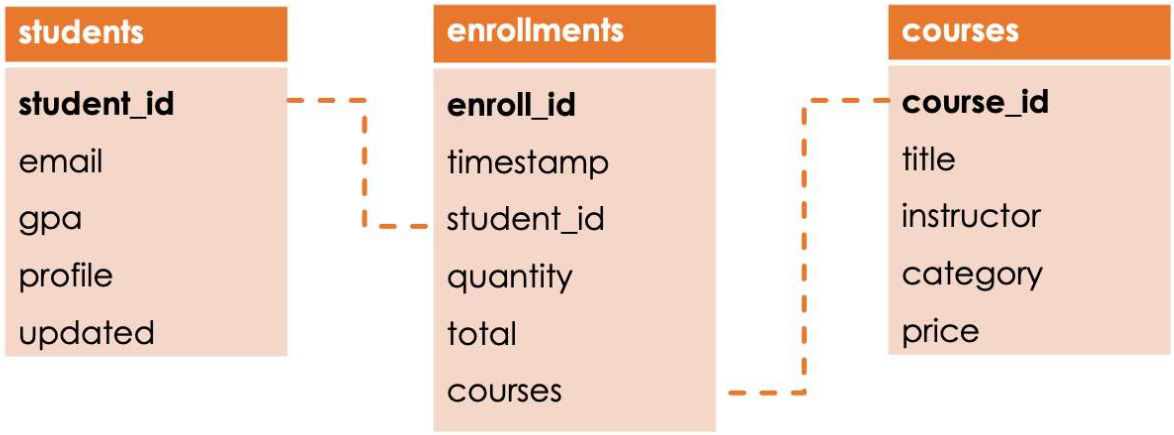
(https://databricks.com)

Lab Solution: Spark Structured Streaming



Run the following cell to setup the lab environment

5

```
%run ../Includes/Setup-Lab
```

Q1- Auto Loader

Use Auto Loader to incrementally load enrollments json files from the directory **{dataset_school}/enrollments-json-raw** into a streaming view called `enrollments_tmp_vw`

7

```
dataset_source = f"{dataset_school}/enrollments-json-raw"
schema_location = "dbfs:/mnt/DE-Associate/checkpoints/school/enrollments_stats"

# ANSWER
(spark.readStream
      .format("cloudFiles")
      .option("cloudFiles.format", "json")
      .option("cloudFiles.schemaLocation", schema_location)
      .load(dataset_source)
      .createOrReplaceTempView("enrollments_tmp_vw"))
```

Q2 - Calculating aggregations on streaming data

Using CTAS syntax, define a new streaming view against `enrollments_tmp_vw` to count the number of enrollments per `student_id`. Name the aggregated field: `enrollments_counts`

9

```
%sql
-- ANSWER
CREATE OR REPLACE TEMPORARY VIEW enrollments_per_student_tmp_vw AS
SELECT
  student_id, count(enroll_id) AS enrollments_count
  FROM enrollments_tmp_vw
  GROUP BY student_id
```

Q3 - Writing stream data

Stream the aggregated data from the `enrollments_per_student_tmp_vw` view to a Delta table called `enrollments_stats`.

```
11
checkpoint_path = "dbfs:/mnt/DE-Associate/checkpoints/school/enrollments_stats"

# ANSWER
query = (spark.table("enrollments_per_student_tmp_vw")
              .writeStream
              .option("checkpointLocation", checkpoint_path)
              .outputMode("complete")
              .table("enrollments_stats")
        )
```

Query the data in the `enrollments_stats` table to ensure data was written as expected.

```
13
%sql
SELECT * FROM enrollments_stats
```

Run the below cell to land a new json file of enrollments data

```
15
load_new_json_data()
```

Verify that the statistics have been updated in the table **enrollments_stats**

```
17
%sql
SELECT * FROM enrollments_stats
```

Q4 - Canceling streaming query

Finally, cancel the above streaming query

```
19
```